

Practice Midterm Exam 2

Curves and surfaces through principal curvatures

75 minutes

Given (first) name: _____

Family (last) name: _____

BU ID: _____

Instructions

- Allow 75 minutes. **Choose any four of the five problems.** Each is worth 20 points, for a total of 80 points. Indicate the omitted problem below; only your four selected problems will be graded.
- In each problem, parts (a)–(c) are computations worth 15 points and part (d) is a proof worth 5 points. Every choice therefore gives **75% computation and 25% proof**.
- Show your calculations and justify your claims. You may use results established in class through principal curvatures, except a result that the problem asks you to prove.
- All parts are stated together. Use the space below each problem and the blank work pages; clearly label work continued elsewhere. Give exact answers. No electronic assistance is needed.
- Dots denote derivatives in an arbitrary parameter t ; primes denote derivatives in arclength s . Write $\kappa \geq 0$ for curvature and k for signed plane curvature.
- Use Kühnel's convention $L = -D\nu \circ (Df)^{-1}$. Thus $L(f_u) = -\nu_u$ and $L(f_v) = -\nu_v$. Principal curvatures are the eigenvalues of L .

Problem omitted (1–5): _____

Problem	1	2	3	4	5	Total
Points possible	20	20	20	20	20	80
Score						

This booklet contains 10 pages: the cover, five problem pages,
and one blank work page between each pair of problems.

Problem 1.**[20 points]**

Consider the space curve

$$c(t) = \left(t, t^2, \frac{2}{3}t^3 \right), \quad t \in \mathbb{R}.$$

- (a) Compute the speed and the arclength function with $s(0) = 0$. Find the length of the segment $0 \leq t \leq 1$. **[5 points]**

- (b) Compute $\kappa(t)$ and $\tau(t)$. You may use

$$\kappa = \frac{\|\dot{c} \times \ddot{c}\|}{\|\dot{c}\|^3}, \quad \tau = \frac{\det(\dot{c}, \ddot{c}, \dddot{c})}{\|\dot{c} \times \ddot{c}\|^2}.$$

[7 points]

- (c) Compute the Frenet frame T, N, B at $t = 0$.

[3 points]

- (d) Let $\gamma(s)$ be any unit-speed Frenet curve on an interval satisfying $\tau(s) = \kappa(s) > 0$. Prove that

$$A(s) = \frac{T(s) + B(s)}{\sqrt{2}}$$

is a constant unit vector, and that T makes an angle $\pi/4$ with this fixed vector. You may use the Frenet equations, but not the slope-line theorem. **[5 points]**

Problem 2.**[20 points]**

Consider the plane curve

$$c(t) = (e^t \cos t, e^t \sin t), \quad t \in \mathbb{R}.$$

Use $J(x, y) = (-y, x)$, so that signed curvature is defined by $dT/ds = kJT$.

- (a) Compute the speed and $s(t)$, with $s(0) = 0$. Find a unit-speed parametrization $\widehat{c}(s)$, including its parameter interval. **[6 points]**

- (b) Compute the signed curvature $k(t)$ and the nonnegative curvature $\kappa(t)$. You may use

$$k(t) = \frac{\det(\dot{c}(t), \ddot{c}(t))}{\|\dot{c}(t)\|^3}.$$

[5 points]

- (c) Compute the signed total curvature of the segment $0 \leq t \leq \pi$, and the length of that segment. **[4 points]**

- (d) Let $\gamma(s)$ be any unit-speed plane curve on an interval with constant signed curvature $k_0 \neq 0$. Prove that its image lies on a circle of radius $1/|k_0|$. *Hint:* Differentiate $\gamma + k_0^{-1}JT$. **[5 points]**

Problem 3.**[20 points]**

Let

$$f(u, v) = \left(u, v, \frac{u^2 + v^2}{2} \right), \quad M = f(\mathbb{R}^2).$$

- (a) Compute f_u, f_v and the upward unit normal. Find the affine tangent plane at $p = f(1, 0)$. **[5 points]**
- (b) Compute the matrix $g = (g_{ij})$ of the first fundamental form for arbitrary (u, v) . At $(1, 0)$, compute the length of $X = f_u + 2f_v$ and the angle between f_u and f_v . **[5 points]**
- (c) Compute the area of the surface region

$$f(\{(u, v) : u^2 + v^2 \leq 1\}).$$

You may use $dA = \sqrt{\det g} \, du \, dv$ and polar coordinates.**[5 points]**

- (d) Prove that $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is an embedding: show that it is an immersion and a homeomorphism onto its image. **[5 points]**

Problem 4.**[20 points]**

Consider the cylinder patch

$$f(u, v) = (2 \cos u, 2 \sin u, u + v), \quad (u, v) \in (-\pi, \pi) \times \mathbb{R},$$

with the outward unit normal.

- (a) Compute f_u, f_v , the unit normal ν , and the matrix of the first fundamental form in the ordered basis (f_u, f_v) . **[5 points]**
- (b) Compute $L(f_u)$ and $L(f_v)$. Express these vectors in the basis (f_u, f_v) , and write the matrix $[L]$ in that basis. **[5 points]**
- (c) Find both principal curvatures and a unit principal direction in $\mathbb{R}^3$ for each. **[5 points]**
- (d) You may use that L is self-adjoint for the first fundamental form I . Prove that if $LX = \lambda X$ and $LY = \mu Y$, with $\lambda \neq \mu$, then $I(X, Y) = 0$. Explain why this does not require the matrix in part (b) to be an ordinary symmetric matrix. **[5 points]**

Problem 5.**[20 points]**

Let

$$f(u, v) = (u, v, u^2 + uv + v^2), \quad p = f(0, 0),$$

and choose the upward unit normal.

- (a) Compute the normal field $\nu(u, v)$. Compute $f_u(0, 0), f_v(0, 0)$ and the first fundamental form matrix at $(0, 0)$. **[5 points]**
- (b) Differentiate ν at $(0, 0)$ and use $L(f_u) = -\nu_u, L(f_v) = -\nu_v$ to find the matrix of L_p in the basis $(f_u(0, 0), f_v(0, 0))$. **[5 points]**
- (c) Find the principal curvatures at p . For each, give a unit principal direction as a vector in $\mathbb{R}^3$. **[5 points]**
- (d) Let $\tilde{f} : U \rightarrow M \subset \mathbb{R}^3$ be any regular surface element, with unit normal $\tilde{\nu}$, and let $p = \tilde{f}(q)$. Prove that

$$D\tilde{\nu}_q(w) \in T_p M \quad \text{for every } w \in \mathbb{R}^2.$$

Explain why $-D\tilde{\nu}_q \circ (D\tilde{f}_q)^{-1}$ therefore maps $T_p M$ to itself. Here the inverse of $D\tilde{f}_q$ is taken on its image $T_p M$. *Hint:* Differentiate $\langle \tilde{\nu}, \tilde{\nu} \rangle = 1$. **[5 points]**

End of exam.